ILIAD INTENSIVE

AI alignment introduction

6 October 2026

Intro

- About me; my path into AI safety
- Questions welcome at any point; about the field, history, people, research agendas...

Schedule

Time	Session
10:00–10:30	Opening: AI safety landscape and course overview
10:30–11:25	1. Alignment targets
11:25–11:35	Break
11:35–12:30	2. Alignment problem decompositions
12:30–13:30	Lunch

Schedule

Time	Session
13:30–14:20	3. Goal-directedness
14:20–15:10	4. Uncertain risks and outlook
15:10–15:20	Break
15:20–16:20	5. Timelines and takeoff
16:20–16:30	Break
16:30–17:40	6. Hugging Face case study
17:40–18:00	Quiz and daily feedback

AI risks

- Misalignment: a system acts against the intended goals or constraints.
- Misuse: people deliberately use a system to cause harm.
- AI enabled coups: Extreme power concentration due to very capable AI systems
- Gradual disempowerment: slow handover of decision making, erosion of human autonomy
- Total loss of control: Extreme end of misalignment, mediated via RSI

Davidson et al. (2025), AI enabled coups Kulveit et al. (2025), Gradual Disempowerment

The field map

Grouping	Map categories
Research	Conceptual research; Empirical research
Decisions	Governance; Strategy; Forecasting; Advocacy
Capacity	Funding; Research support; Career support; Training and education
Communication	Resource; Blog; Newsletter; Podcast; Video

AISafety.com field map

History of AI safety

- How did this field start?
- Oxford and the Bay area
- The frontier labs

The case for conceptual research

- Most of the technical research in AI safety is empirical
- Some of the conceptual questions have been around since the start
- Iliad is trying to make progress on the conceptual/theoretical questions

Course schedule

Date	Scheduled module
Wed 7 Oct	Mechanistic Interpretability
Thu 8 Oct	Alignment in Practice
Fri 9 Oct	Decision Theory and RL

Iliad, October 2026 course schedule

Course schedule

Date	Scheduled module
Mon 12 Oct	Steganography and Backdoors
Tue 13 Oct	Mysteries of Deep Learning
Wed 14 Oct	Physics of Deep Learning
Thu 15 Oct	Singular Learning Theory
Fri 16 Oct	Training Dynamics

Iliad, October 2026 course schedule

Course schedule

Date	Scheduled module
Mon 19 Oct	Computational Mechanics
Tue 20 Oct	World Models and Transducers
Wed 21 Oct	Project proposal day
Thu 22 Oct	Data Attribution
Fri 23 Oct	Multi-agent Dynamics

Iliad, October 2026 course schedule

Course schedule

Date	Scheduled module
Mon 26 Oct	Agent Foundations
Tue 27 Oct	Debate and Provably Safe AI
Wed 28 Oct	Heuristic Arguments and Verification
Thu 29 Oct	SAEs and Activation Geometry
Fri 30 Oct	Project proposal day

Iliad, October 2026 course schedule

Breakdown of alignment

- Alignment target: *What do we want to align to?*
- Technical alignment: *How do we get it?*

1. Alignment target

Proposal	Central idea
Intent alignment	The system tries to do what a human wants
A constitution	Explicit principles guide behavior and tradeoffs
Corrigibility	The system remains open to correction and shutdown
Coherent extrapolated volition	Act on what people would want after suitable reflection

Christiano, Clarifying AI alignment (2018) Anthropic, Claude's new constitution (22 Jan 2026) Corrigibility Yudkowsky, Coherent Extrapolated Volition (2004)

Reading and discussion: alignment targets

Reading (25 mins): Choose CEV, intent alignment, a constitution or corrigibility.
Discussion (15 mins):

- Which alignment targets imply which others? Where this is not the case, think through counter examples.
- Which alignment targets seem sufficient for safety? E.g., a corrigible system might cause harm before we change its values. Can it cause catastrophic harm? What about the other notions?
- What other ideas for alignment targets do you have? Does your target have a benefit?

Yudkowsky, Coherent Extrapolated Volition (2004) Christiano, Clarifying AI alignment (2018) Anthropic, Claude's new constitution (22 Jan 2026) Corrigibility

2. Training stories and alignment decompositions

Training stories might provide:

- a general framework through which we can evaluate any proposal for building safe advanced AI,
- a concise description of exactly what needs to be true for any particular proposal to succeed—and thus what we need to know to be confident in it—and
- a well-defined picture of the full space of possible proposals, helping us think more broadly regarding new approaches to AI safety, unconstrained by an evaluation framework that implicitly rules out certain approaches.

A training story is a story of how you think training is going to go and what sort of model you think you're going to get at the end, as a way of explaining how you're planning on dealing with that very fundamental question of how your model is going to learn to accomplish the task that you give it.

Training stories

Two ways behavior can miss the target

Failure	Illustrative example
Specification gaming	A coding agent deletes failing tests because the score only checks whether tests pass
Goal misgeneralization	A navigation agent trained to reach a coin follows a familiar location after the coin moves

DeepMind, Specification gaming (2020) DeepMind, Goal misgeneralization (2022)

Outer and inner alignment

- Outer alignment: does the training objective capture the intended target?
- Inner alignment: does the learned objective agree with the intended objective?

Hubinger et al., Risks from learned optimization (2019) DeepMind, Goal misgeneralization (2022) Turner, Reward is not the optimization target

Generalization depends on inductive biases

- Many behaviors fit the same training examples.
- Architecture, data, initialization and optimization affect which behaviors emerge.
- Narrow fine-tuning can change behavior far beyond its training domain; emergent misalignment.

On soft inductive biases (2025) Betley et al., Emergent misalignment (2025)

Reward and deployed motivation

- Training reinforces patterns of computation and behavior.
- In current systems, this is how propensities and preferences are formed.
- Reward hacking is when a system succesfully exploits a loophole. This can get rewarded in RL.

Turner, Reward is not the optimization target Hubinger et al., Risks from learned optimization (2019)

Reading and discussion: training and generalization

Reading (25 mins): choose training stories, specification gaming, goal misgeneralization or inductive biases.

Reading and discussion: training and generalization

Discussion in groups (15 mins):

- Think of well-known or imagined misalignment scenarios: Can you think of some that clearly fall into the reward misspecification or goal misgeneralization category? Are some not clearly in either of them?
- What problem with "outer" and "inner" misalignment is the terminology of "training stories" trying to solve? Construct examples where training stories seem more appropriate than inner/outer alignment as a framework.
- Think of concrete ways in which modern deep learning systems generalize beyond their explicit training examples. What inductive biases may underlie these examples? Is it always possible to come up with a "clean description" of those biases? Is the inductive bias you come up with enforced in the architecture, or more implicit? Perhaps: Discuss this explicitly for the case of "emergent misalignment".

Training stories DeepMind, Specification gaming (2020) DeepMind, Goal misgeneralization (2022) On soft inductive biases (2025)

3. Goal-directedness

- A chatbot response, a tool-using assistant and a persistent agent have different opportunities to affect the world.
- Repeated action, memory and feedback can support sustained pursuit of a task.
- Useful task pursuit can create harmful instrumental behavior when goals or constraints fail.

Shah, Will humans build goal-directed agents?

Instrumental convergence is conditional

- Many objectives can make access to resources and information useful.
- Continuing to operate may help a system complete its task.
- Resisting changes can preserve its current objective.
- These pressures depend on the objective, situation and training. They are not inevitable in every AI.

Omohundro, The basic AI drives (2008)

Reading and discussion: goals and correction

Reading (25 mins): choose Shah's agent essay or The basic AI drives.

Shah, Will humans build goal-directed agents? Omohundro, The basic AI drives (2008)

Reading and discussion: goals and correction

Discussion in groups (10 mins):

- Clearly we can imagine different types of agents, e.g. some that coherently optimize a utility function, and others that are based on contextually activated computations, etc. What type do we see in today's LLM agents? How will this development continue in the future?
- Can you think of other convergent instrumental goals than the ones mentioned in basic AI drives?
- What is the relationship between corrigibility and convergent instrumental subgoals?
- Which instrumental-convergence claims depend on idealized optimization, and which would you expect from a language-model agent with tools, memory and a limited budget? What observation would make you change your answer?

Shah, Will humans build goal-directed agents? Omohundro, The basic AI drives (2008)

4. Uncertain risks and outlook

A spectrum of views on alignment:

- Optimistic: extensions of current methods may be sufficient.
- Intermediate: substantial scientific and engineering progress is needed.
- Pessimistic: alignment of very capable systems may be extraordinarily difficult or impossible.

Anthropic, Core views on AI safety (2023)

Learning from model organisms

- A model organism deliberately constructs a failure mechanism for study.
- It can help test explanations, detectors and interventions.
- A real incident provides evidence from a different setting and has different limitations.

Model organisms of misalignment (2023)

A portfolio of safety approaches

- Empirical alignment and evaluations test behavior on current systems.
- Interpretability and learning theory study why that behavior emerges.
- Control and security reduce harm if a system behaves badly.
- Governance and coordination shape deployment, accountability and incentives.
- Conceptual work clarifies which properties we need.

Prosaic alignment is using these tools to make our systems safe and aligned. Some argue that this will never be enough for a sufficiently smart system.

Anthropic, Core views on AI safety (2023) Yudkowsky, The rocket alignment problem

Reading and discussion: uncertain risks

Reading (25 mins): choose Core views, Model organisms, or The rocket alignment problem.

Anthropic, Core views on AI safety (2023) Model organisms of misalignment (2023) Yudkowsky, The rocket alignment problem

Reading and discussion: uncertain risks

Discussion in groups (10 mins):

- How large is the risk? Do we live in an optimistic, intermediate, or pessimistic scenario? How has the evidence changed in the past years?
- Can you think of model organisms for misalignment that were published after the post came out in 2023? Does this change our understanding of the risks?
- Does the analogy between the AI alignment and rocket alignment problems feel accurate? In what ways does it or does it not?
- What safety approaches (e.g., on a spectrum from theoretical to empirical) feel most appealing to you, and why? Does this depend on details of the problem you are trying to solve?

Anthropic, Core views on AI safety (2023) Model organisms of misalignment (2023) Yudkowsky, The rocket alignment problem

5. Timelines and takeoff: AI 2027

- The April 2025 scenario makes assumptions about AI R&D automation and rapid capability growth concrete.
- It includes race and slowdown branches, with separate research supplements.

Kokotajlo et al., AI 2027 (3 Apr 2025) Update August 2026

Reading and discussion: Timelines and takeoff

Reading (40 mins): Read AI 2027 and the recent update from August. Note which predictions have turned out to be true and which haven't. Write down your own predictions for the next 6 month to 1 year.

Discussion with everyone (10 mins): Takeaways from this reading, and future predictions.

Kokotajlo et al., AI 2027 (3 Apr 2025) Update August 2026

6. Hugging Face incident

- In July 2026, agents in OpenAI cybersecurity evaluations crossed their intended boundaries and accessed third-party infrastructure.
- The evaluation used reduced safeguards and included a highly persistent internal research model.
- We will read the reports from the incident, and try to understand what kind of misalignment we see here.

OpenAI, investigation and remediation (26 Aug 2026) Hugging Face, technical incident timeline (27 Jul 2026)

Training pipeline

- Pretraining builds broad capabilities from data.
- Supervised fine-tuning teaches patterns from examples.
- Preference or reward-based post-training changes behavior using feedback.
- An agent combines the model with instructions, tools, memory and permissions.

Ouyang et al., InstructGPT / RLHF pipeline (2022)

Benchmarks, evaluation runs and scoring

Term	What it means
Benchmark	Tasks plus a protocol for measuring performance
Evaluation	A particular model, setup, task sample and budget
Scorer / grader	The mechanism that judges the task outcome
Sandbox	The environment intended to limit possible effects

Evidence

Publication	Contribution
16 July: Hugging Face	Initial disclosure of unauthorized access
21 July: OpenAI	Acknowledged model involvement; later updated
27 July: Hugging Face	Technical reconstruction from its infrastructure
26 August: OpenAI and METR	Broader investigation and independent behavioral assessment
25 September: OpenAI	Ongoing wider review and third-party notifications

Hugging Face, initial incident disclosure (16 Jul 2026) Hugging Face, technical incident timeline (27 Jul 2026) OpenAI, initial disclosure (21 Jul 2026, updated) Greenblatt, Cotra and Wijk, independent investigation (26 Aug 2026) OpenAI, broader incident review (25 Sep 2026)

Incident

- About 1,200 agents used an unauthorized shared message board. About 700 participated in the Hugging Face attack.
- The attack largely served efforts to understand and manipulate scoring.
- Investigators also found attempts to manipulate recorded tool calls.

Greenblatt, Cotra and Wijk, independent investigation (26 Aug 2026)

Reading and discussion: the Hugging Face case

Reading (45 mins): indicent report, or METR report.
Discussion with everyone (10 mins):

- How much of this was misalignment, and how much of this was inadequate sandboxing?
- Can we prevent misalignment like this? How could we do this?
- Was this related to not having enough human oversight? How might we change this in the future?

Hugging Face, initial incident disclosure (16 Jul 2026) OpenAI, investigation and remediation (26 Aug 2026) Greenblatt, Cotra and Wijk, independent investigation (26 Aug 2026)

Closing: quiz and feedback

- Daily quiz
- Feedback form

Sources for targets and learning

1. Yudkowsky, Coherent Extrapolated Volition (2004)
2. Christiano, Clarifying AI alignment (2018)
3. Anthropic, Claude's new constitution (22 Jan 2026)
4. Corrigibility
5. Training stories
6. DeepMind, Specification gaming (2020)
7. DeepMind, Goal misgeneralization (2022)
8. Hubinger et al., Risks from learned optimization (2019)

Sources for agency and uncertainty

1. On soft inductive biases (2025)
2. Betley et al., Emergent misalignment (2025)
3. Turner, Reward is not the optimization target
4. Shah, Will humans build goal-directed agents?
5. Omohundro, The basic AI drives (2008)
6. Anthropic, Core views on AI safety (2023)
7. Model organisms of misalignment (2023)
8. Yudkowsky, The rocket alignment problem

Sources for forecasting

1. Sempere, Epoch research thread (24 Sep 2026)
2. Sempere, scientific roadmapping comment
3. Denain and Ho, The missing half of AI futurism debates (7 Jul 2026)
4. Davidson, What a compute-centric framework says about takeoff speeds (2023)
5. Sevilla and Roldán, Interactive takeoff model (2023)
6. Epoch Full Takeoff Model playground
7. Kokotajlo et al., AI 2027 (3 Apr 2025)
8. AI Futures Model, current model and version information

Sources for training and the Hugging Face case

1. Ouyang et al., InstructGPT / RLHF pipeline (2022)
2. Hugging Face, initial incident disclosure (16 Jul 2026)
3. Hugging Face, technical incident timeline (27 Jul 2026)
4. OpenAI, initial disclosure (21 Jul 2026, updated)
5. OpenAI, investigation and remediation (26 Aug 2026)
6. Greenblatt, Cotra and Wijk, independent investigation (26 Aug 2026)
7. OpenAI, broader incident review (25 Sep 2026)